A

Mini Project Report

on

APSIT CONNECT

Submitted in partial fulfilment of the requirements for the degree

Second Year Engineering – Computer Science and Engineering (Data Science)

by

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Academic Year: 2025–26

CERTIFICATE

This is to certify that the Mini Project report on APSIT CONNECT has been submitted by Aryan Salunke (24107115), Kartik Nair (23107123), Siddhant Tidke (23107083), and Gautam Mandal (24107109), who are bonafide students of A. P. Shah Institute of Technology, Thane, as a partial fulfillment of the requirement for the degree in **Computer Science Engineering (Data Science)**, during the academic year **2025–2026** in the satisfactory manner as per the curriculum laid down by University of Mumbai.

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ACKNOWLEDGEMENT

This project would not have come to fruition without the invaluable help of our guide Ms. Ujwala Pagare. Expressing gratitude towards our HoD, Dr. Pravin Adivarekar, and the Department of Computer Science Engineering (Data Science) for providing us with the opportunity as well as the support required to pursue this project. We would also like to thank our project coordinators Prof. Richa Singh and Prof. Kiran Gaikwad who gave us valuable suggestions and ideas when we were in need of them. We would also like to thank our peers for their helpful suggestions.

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Chapter 1

Introduction

APSIT Connect is a centralized academic platform designed to simplify and enhance the way students and faculty interact, share resources, and manage academic activities within an institution. In many colleges, academic communication is often scattered across multiple platforms such as WhatsApp, email, and informal sharing methods. This leads to issues such as missed updates, loss of important files, lack of proper organization, and inefficient communication between students and faculty. As a result, students often struggle to access notes, assignments, and project resources in a structured and reliable manner.

The aim of this project is to develop a unified and user-friendly system that brings all academic functionalities into a single platform. APSIT Connect provides a secure login and registration system that ensures only verified students and teachers can access the platform. Once logged in, users are directed to role-based dashboards tailored to their needs. Students can access study materials, view attendance, submit assignments, and participate in academic discussions. Teachers, on the other hand, can upload notes, manage attendance, review assignments, and guide students through discussion forums.

The system is designed with a focus on organization and accessibility. Academic content is categorized subject-wise and year-wise, making it easier for users to search and retrieve information quickly. The discussion module serves as an interactive space where students can ask doubts and receive guidance from faculty members, thereby improving engagement and understanding. By digitizing these processes, APSIT Connect reduces dependency on manual methods and scattered communication tools.

This project not only addresses the existing problems in academic resource management but also contributes to improving overall efficiency, transparency, and collaboration within the institution. It demonstrates how modern technologies can be effectively used to build a structured academic ecosystem that benefits both students and faculty. The scope of the system can be extended in the future by integrating cloud storage, real-time notifications, and advanced analytics for better academic tracking and performance evaluation.

1.1 Purpose

The purpose of this project is to design and develop a centralized academic platform named APSIT Connect that simplifies the management of academic activities and improves communication between students and faculty. The system aims to provide a secure and user-friendly environment where only verified users can access the platform through a login and registration system. It integrates several key features such as role-based dashboards for students and teachers, allowing personalized access to functionalities based on user roles. The platform enables students to upload and download academic materials like notes, assignments, and project files, while teachers can manage and share resources efficiently.

Additionally, APSIT Connect includes subject-wise and year-wise organization of content to ensure easy searching and structured storage of academic data. The system also provides a discussion module where students can post their doubts and receive guidance from faculty, thereby enhancing interaction and learning. Features such as attendance tracking, assignment submission, and resource sharing are incorporated to replace traditional manual methods with a digital solution. Overall, the purpose of this project is to create a unified platform that enhances accessibility, organization, and communication within the academic environment.

1.2 Problem Statement

In many educational institutions, academic communication and resource sharing are not managed through a centralized system. Students often rely on platforms like messaging applications or emails to receive notes, assignments, and important updates. This leads to several issues such as missed information, difficulty in organizing study materials, and lack of proper record maintenance. Additionally, communication between students and faculty for doubt-solving is often unstructured and inconsistent. There is also no efficient system for managing attendance, assignment submissions, and academic discussions in one place. These limitations create inefficiencies in the learning process and reduce overall academic productivity. Therefore, there is a need for a unified system that can streamline these activities and provide a more organized academic environment.

1.3 Objectives

APSIT Connect is designed to improve the academic experience by providing a structured and efficient digital platform for students and faculty. The primary objective of this project is to develop a centralized system that simplifies access to academic resources and enhances communication within the institution. Each feature of the system is designed to fulfill a specific purpose and improve usability.

1.Secure Login and Registration System

The system ensures that only authorized users such as students and faculty can access the platform through a secure authentication mechanism. This helps maintain data privacy and restricts access to verified users only.

2. Centralized Resource Sharing

APSIT Connect allows users to upload and download academic materials such as notes, assignments, and project files. This ensures that all resources are available in one place, reducing dependency on multiple platforms.

3. Organized Content Management

The system organizes academic content subject-wise and year-wise, making it easier for users to search and access required materials efficiently without confusion.

4. Discussion and Interaction Module

The platform provides a discussion section where students can post their doubts and faculty members can respond. This improves communication and helps in better understanding of academic concepts.

5. Academic Activity Management

Features like attendance tracking and assignment submission are included to digitize traditional processes, making them more efficient and easier to manage.

1.4 Scope

APSIT Connect aims to serve as a comprehensive academic platform that caters to a wide range of users within an educational institution. The scope of this project includes providing a centralized system for academic resource sharing, communication, and management of academic activities.

1. Students and Learners

The system is highly beneficial for students who need easy access to study materials, assignments, and project resources. It helps them stay organized and improves their learning experience.

2. Faculty Members

Teachers can use the platform to upload study materials, manage attendance, review assignments, and guide students through discussions. This reduces manual effort and improves efficiency.

3. Academic Institutions

The platform can be implemented in colleges to streamline academic processes and improve coordination between students and faculty. It supports digital transformation in education.

4. Collaborative Learning Environment

APSIT Connect promotes collaboration by allowing students to interact, share resources, and participate in discussions, creating a more engaging learning environment.

5. Future Expansion

The system has the potential to be extended with advanced features such as cloud storage, realtime notifications, analytics, and integration with college ERP systems to enhance function

Chapter 2

Proposed System

APSIT Connect is a centralized platform where students can upload and download academic resources easily. It also provides a discussion forum where students can ask doubts and receive guidance from faculty members. The system is built using Flutter and Dart with a role-based access model.

2.1 Features and Functionality

Feature 1: Login System

Allows users to log in using email and password with role selection (Student or Teacher) for secure access. The system validates credentials and routes the user to the appropriate dashboard based on their role.

Feature 2: Role-Based Dashboard

Displays a personalized dashboard with modules such as Attendance, Notes, Discussion, and Assignments according to the logged-in user role. Students see their learning resources while teachers see management tools.

Feature 3: Attendance Module

Provides a section where students can view their attendance records. Teachers can manage, update, and track student attendance through a dedicated interface.

Feature 4: Discussion Module

Provides a platform where students can post academic doubts and queries. Teachers can respond to student questions, enabling organized and accessible communication for the entire class.

Feature 5: Notes and Resource Sharing

Students can upload and download academic materials such as notes, assignments, and project files. Content is organized subject-wise and year-wise for easy navigation.

Chapter 3

Project Outcomes

The APSIT Connect project successfully achieves its goal of providing a centralized and efficient academic platform for students and faculty. The system integrates multiple functionalities into a single application, improving accessibility, communication, and organization within the academic environment. The key outcomes of the project are described below:

1. **User-Friendly Interface**

   The system is designed with a clean and intuitive user interface that ensures ease of use for both students and faculty. The dashboard-based layout allows users to navigate between modules such as attendance, notes, assignments, and discussion without confusion. The use of structured layouts like GridView enhances the overall user experience and makes the application visually organized.

2. **Centralized Academic Resource Management**

   APSIT Connect provides a single platform where all academic materials such as notes, assignments, and project files can be uploaded, accessed, and managed efficiently. This eliminates the need for multiple communication channels and ensures that all resources are available in an organized and accessible format.

3. **Role-Based Access System**

   The project successfully implements a role-based login system that differentiates between student and teacher functionalities. Students can access learning materials and submit assignments, while teachers can upload resources, manage attendance, and respond to student queries. This separation ensures better control and functionality within the system.

4. **Improved Communication and Interaction**

   The discussion module enables effective communication between students and faculty. Students can post doubts and questions, and teachers can provide guidance in a structured manner. This feature enhances engagement and promotes collaborative learning within the academic environment.

5. **Efficient Academic Activity Management**

The platform digitizes traditional academic processes such as attendance tracking and assignment submission. Teachers can easily manage records, and students can view their progress, reducing manual effort and improving efficiency.

6. **Structured Data Organization**

Academic content is organized subject-wise and year-wise, making it easier for users to search and retrieve information. This structured approach reduces confusion and improves accessibility of study materials.

7. **System Performance and Reliability**

The application performs efficiently using Flutter and Dart, providing smooth navigation and quick response times. The use of local in-memory data handling ensures fast access during runtime, making the prototype responsive and reliable.

8. **Scope for Future Enhancement**

The project is designed with scalability in mind. It can be extended with advanced features such as cloud storage, real-time notifications, database integration, and analytics. These enhancements can further improve functionality and make the system suitable for real-world deployment.

Chapter 4

Software Requirements

The development of APSIT Connect requires a combination of frontend technologies, backend/data handling mechanisms, and development tools to ensure efficient performance and usability. The software requirements are categorized as follows:

Frontend

1. **Flutter Framework**

   Flutter is used to design and develop the user interface of the application. It provides a rich set of widgets that help in creating responsive and visually appealing mobile applications.

2. **Dart Programming Language**

   Dart is used for implementing the application logic and handling user interactions. It works seamlessly with Flutter and ensures smooth performance

3. **Material Design (Material UI)**

   Material UI is used to maintain consistency in design and provide modern UI components such as buttons, cards, and input fields

4. **Grid View Layout**

   Grid View is used to organize dashboard modules like Attendance, Notes, Discussion, and Assignments in a structured manner.

5. **Navigator**

   Navigator is used for managing screen transitions and enabling smooth navigation between different pages of the application.

Backend

1. Local In-Memory Database

A Map data structure is used to store user credentials and application data during runtime for quick access and processing

2. Role-Based Authentication

The system ensures secure login by allowing access based on user roles such as Student and Teacher.

3. Data Handling using Dart

Dart is used to manage data operations such as storing user input, retrieving academic content, and controlling access to modules.

4. Session Management

Maintains user session after login to ensure continuous access without repeated authentication.

Development Tools

1. Flutter SDK

Used for building, testing, and running the application.

2. Visual Studio Code

These IDEs are used for writing code, debugging, and managing the project efficiently.

3. Android Emulator

Used for testing the application on different screen sizes and environments.

Chapter 5

Project Design

The system architecture of APSIT Connect is illustrated using a layered block diagram, as shown in Figure 5.1, which represents the overall structure and flow of the application. The system is divided into multiple layers, each responsible for specific functionalities, ensuring modularity, scalability, and efficient operation.

At the top layer, Layer 1 – User Roles, defines the different types of users interacting with the system, namely Admin, Teacher, and Student. Each role has specific functionalities assigned to it. The admin is responsible for managing users and courses, the teacher can upload notes, assign tasks, and manage attendance, while the student can access study materials, submit assignments, and view attendance records. This role-based structure ensures controlled access and personalized interaction with the system.

The next layer, Layer 2 – Flutter Application, represents the main application interface developed using Flutter and Dart. This layer includes core modules such as authentication and login, course management, user management, notes viewing, assignments, attendance tracking, discussion forum, and timetable management. These modules provide all the essential features required for academic activities and are designed to offer a smooth and user-friendly experience.

Below this, Layer 3 – Backend Services, consists of Firebase services that handle data storage and authentication. Firebase Authentication is used for secure login and session management, Cloud Firestore is used for storing structured data such as user details, courses, and attendance records, and Firebase Storage is used for storing files such as notes and assignments. This backend layer ensures efficient data handling and real-time access to information.

The bottom layer, Layer 4 – Platform and Technology Stack, includes the technologies used to build and deploy the system, such as Flutter SDK, Dart programming language, Android APK, iOS IPA,

and Firebase services. This layer highlights the tools and frameworks that support the overall functioning of the application.

Additionally, the diagram also represents the data flow within the system, where user actions are processed through the Flutter UI, authenticated via Firebase Authentication, stored in Firestore database, and reflected back to the user interface. As shown in Figure 5.1, this layered architecture ensures proper separation of concerns, efficient data flow, and smooth interaction between different components of the system.

Overall, the block diagram clearly demonstrates how APSIT Connect is structured into different layers, enabling organized development, easy maintenance, and scalability for future enhancements.

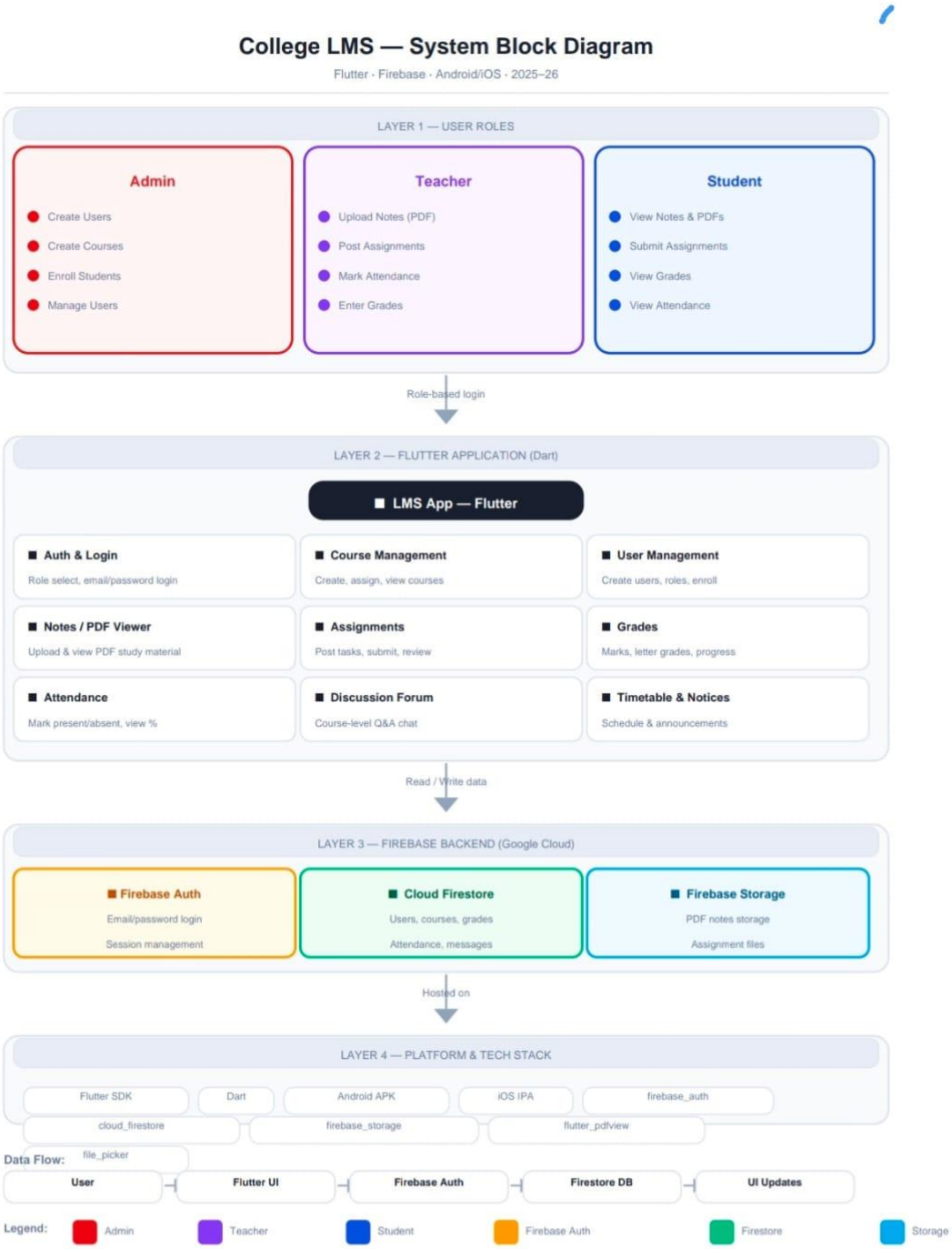

Figure 5.1: Block Diagram of APSIT Connect

Chapter 6

Project Scheduling

The work distribution of the APSIT Connect project, as represented in Figure 6.1, was carried out in a systematic and well-planned manner over the defined timeline from January 13 to March 31, 2026. In the initial phase, all team members collaboratively worked on group formation, topic finalization, and defining the scope and objectives of the project during the third and fourth weeks of January. This phase also included problem identification, feasibility study, and requirement gathering, which helped in understanding the core functionalities of the system.

In the next stage, during the end of January and the first week of February, Aryan Salunke focused on requirement analysis and system architecture design, while Siddhant Tidke worked on user interface design and module planning. This ensured that both the structure and design of the system were clearly defined before moving to the development phase.

During the development phase from the first to the third week of February, Aryan Salunke and Siddhant Tidk**e** worked on frontend development using Flutter, designing dashboards and implementing modules such as attendance, notes, and discussion. At the same time, Kartik Nair handled backend logic and data management using Dart, including role-based authentication and integration of modules. The integration process continued towards the end of February and extended to the first week of March, ensuring smooth communication between all components of the system.

In the following phase during March, Siddhant Tidke worked on testing activities, including unit testing, system testing, and bug fixing, to ensure the system functioned correctly without errors. Meanwhile, Gautam Mandal focused on documentation, report preparation, and organizing the final project structure.

Finally, all team members collaborated in the last week of March for final integration, testing, and submission of the project. The Gantt chart clearly reflects this timeline and task allocation, as shown in Figure 6.1, ensuring efficient coordination, proper distribution of responsibilities, and successful completion of APSIT Connect within the planned schedule.

GANTT CHART

Project Title : APSIT CONNECT – Academic Platform
Project Duration : January 13, 2026 to March 31, 2026

Prepared By: [Your Name]
Guide: [Guide Name]

S.No.	Project Activities / Tasks	Start Date	End Date	Duration (Days)
Phase 1: Planning & Requirement Analysis				
1	Problem Identification	13-01	15-01	3
2	Feasibility Study	15-01	18-01	4
3	Requirement Gathering	18-01	22-01	5
4	Requirement Analysis	22-01	28-01	7
5	Documentation of Requirements	28-01	30-01	3
Phase 2: System Design				
6	System Architecture Design	30-01	05-02	7
7	Database Design	05-02	10-02	6
8	UI/UX Design	10-02	18-02	9
9	Module Design	18-02	25-02	8
Phase 3: Development				
10	Frontend Development	01-02	22-02	22
11	Backend Development	05-02	28-02	24
12	Integration of Modules	20-02	05-03	14
Phase 4: Testing				
13	Unit Testing	06-03	12-03	7
14	System Testing	12-03	20-03	9
15	Bug Fixing	20-03	25-03	6
Phase 5: Deployment & Documentation				
16	User Documentation	10-03	20-03	11
17	Final Testing & Review	20-03	25-03	6
18	Final Report Preparation	25-03	30-03	6
19	Project Submission	31-03	31-03	1
Total Duration (Days)			**78 Days**	

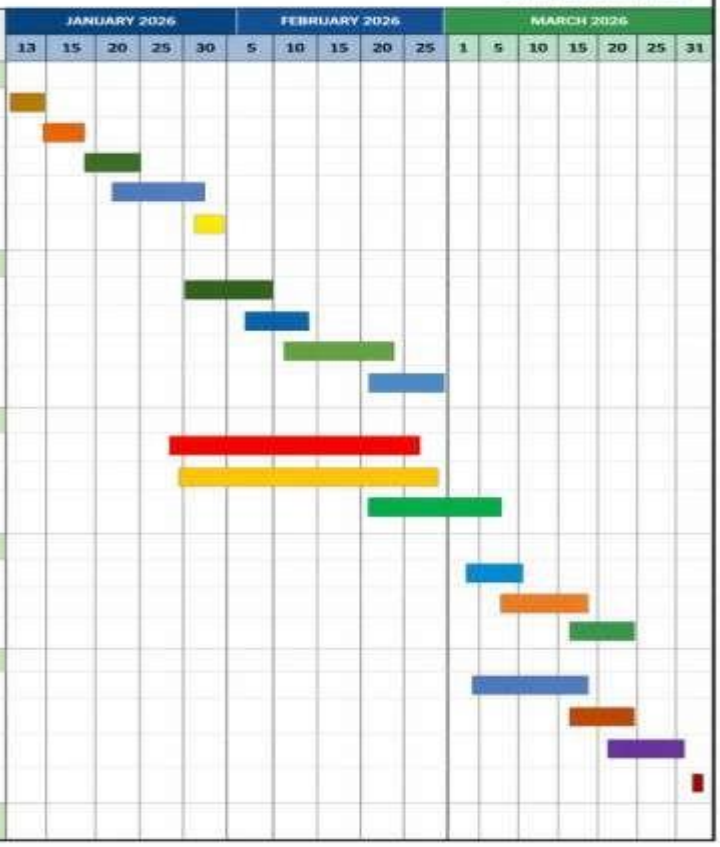

Figure 6.1: Project Gantt Chart (January 13 – March 31, 2026)

Chapter 7

Results

The following results were observed after developing and testing APSIT Connect:

7.1 Login Screen

The login screen serves as the entry point of the APSIT Connect application and ensures secure access to the system. It provides role-based authentication, allowing users to log in as Admin, Teacher, or Student. Users select their role and enter their email and password, after which the system validates the credentials and grants access accordingly. Upon successful login, users are redirected to their respective dashboards based on their roles. As shown in Figure 7.1, the login interface is simple, secure, and user-friendly, ensuring proper access control and data security within the system.

Figure 7.1: Login Screen

7.2 Admin Dashboard

The admin dashboard provides a centralized control panel for managing the overall functionality of the APSIT Connect application. It allows administrators to manage users, courses, and system data efficiently. The admin can add, update, or remove users, monitor academic activities, and ensure proper system functioning. As illustrated in Figure 7.2, the dashboard presents all administrative features in a structured layout, enabling smooth navigation and efficient execution of management tasks. This enhances system control and maintains data integrity.

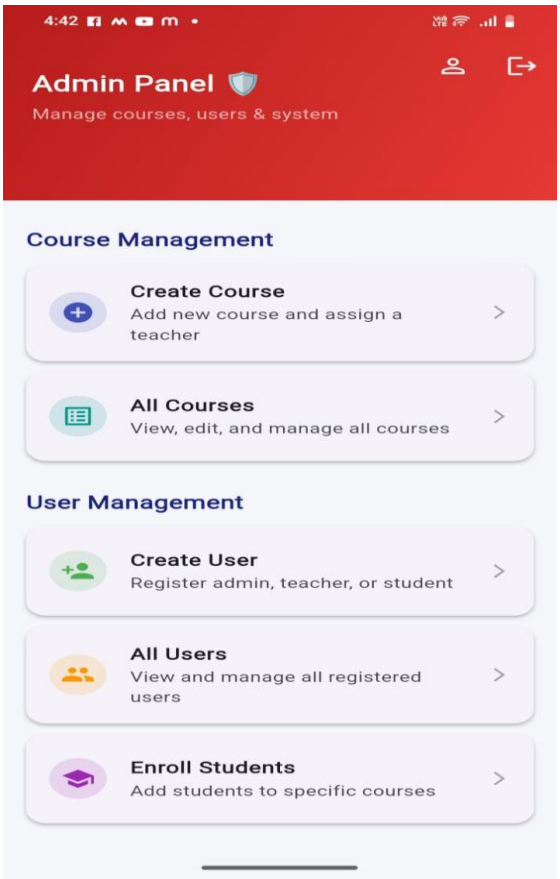

Fig 7.2 Admin Panel

7.3 Student Dashboard

The student dashboard provides a centralized interface for accessing academic functionalities available to students. It includes modules such as attendance viewing, notes access, assignment submission, and discussion forum. The dashboard is designed using a GridView layout, which organizes features into clearly visible sections for easy navigation. As shown in Figure 7.3, students can quickly access different modules from a single screen, improving usability and efficiency. This structured design helps students manage their academic activities effectively.

Fig 7.3 Student Dashboard

7.4 Teacher Dashboard

The teacher dashboard is designed to provide faculty members with tools to manage academic activities efficiently. It includes features such as attendance management, uploading study materials, reviewing assignments, and responding to student queries. As illustrated in Figure 7.4, the dashboard provides easy access to all required modules, allowing teachers to perform multiple tasks from a single interface. This improves efficiency and simplifies academic management for faculty members.

Fig 7.4 Teacher Dashboard

7.5 Discussion Module

The discussion module serves as an interactive platform that facilitates communication between students and teachers. Students can post their doubts or questions, and teachers can respond to them, creating a structured discussion thread. As shown in Figure 7.5, the interface displays questions and responses in an organized manner, making it easy to follow discussions. This module enhances collaboration and promote better understanding of academic concepts.

Fig 7.5 Discussion Screen

7.6 Attendance Module

The attendance module is used to manage and track student attendance digitally. Students can view their attendance records, while teachers can mark and update attendance. As illustrated in Figure 7.6, the module provides a clear and structured view of attendance data, enabling users to monitor academic performance efficiently. This replaces manual attendance systems and improves accuracy and transparency.

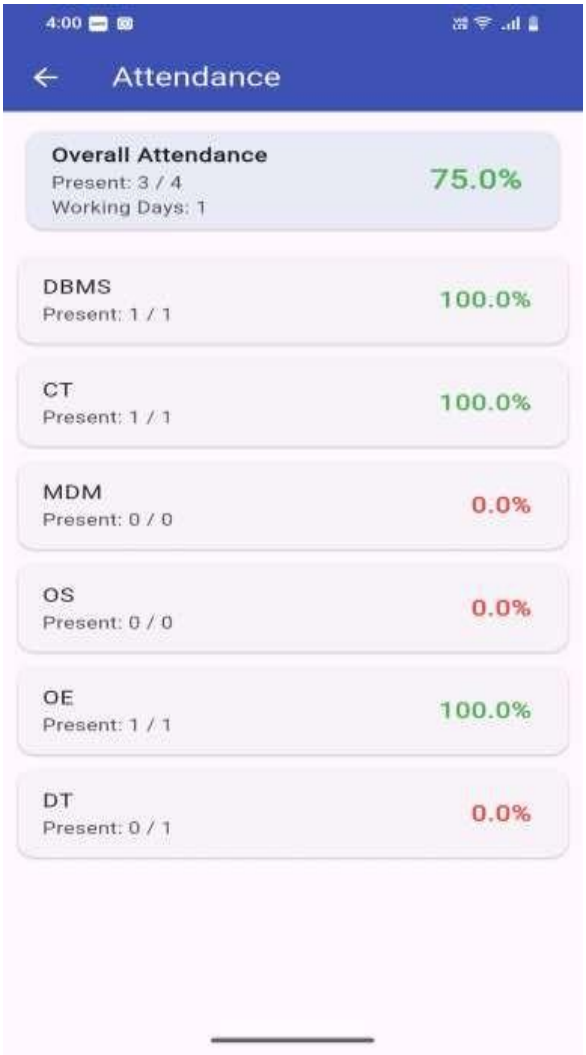

Fig 7.6 Attendance Screen

Chapter 8

Conclusion

APSIT Connect is a centralized academic platform developed to address the challenges of fragmented communication and resource sharing in educational institutions. The system successfully integrates multiple academic functionalities—such as secure login, role-based dashboards, resource sharing, attendance management, and a discussion forum—into a single, user-friendly application. By bringing these features together, the platform reduces dependency on scattered tools and provides a structured environment for both students and faculty.

A key strength of the project is its role-based design, which tailors functionality according to user type. Students can easily access notes, submit assignments, view attendance, and participate in discussions, while teachers can upload materials, manage attendance, review submissions, and guide students. This separation improves usability and ensures that each user interacts only with relevant features. The subject-wise and year-wise organization of content further enhances accessibility, making it easier to locate and manage academic resources.

The system also improves communication and collaboration through the discussion module, where students can post doubts and receive timely guidance from faculty. Additionally, digitizing routine processes like attendance and assignment handling increases efficiency, reduces manual errors, and maintains better records. The use of Flutter and Dart enables a responsive interface with smooth navigation and reliable performance in the prototype.

From a design perspective, APSIT Connect follows a modular and scalable architecture, allowing future enhancements without disrupting existing functionality. The project can be extended with features such as cloud-based storage, real-time notifications, integration with institutional ERP systems, and analytics for tracking student performance. These improvements can transform the prototype into a fully deployable, institution-wide solution.

In conclusion, APSIT Connect demonstrates how a well-designed application can streamline academic workflows, enhance communication, and improve overall productivity within an educational environment. It provides a practical foundation for digital transformation in academia and can be further evolved to meet the growing needs of modern institutions.

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